Chapter 18
Integrated Approach for Characterization of Highly Heterogeneous Drugs



Rahul Raman, Zachary Shriver, Thomas Clark and Ram Sasisekharan

Abstract The transformative evolution of the life sciences field is at an important point of convergence with data sciences, which is another dramatically evolving field. The advances in rapid, sensitive, and high-throughput analytical methods and "big data" approaches (deep sequencing, etc.) have led to an explosion of datasets in life sciences. This rapid accumulation of diverse data necessitates the development of scientific frameworks and concepts borrowing from the evolving data sciences field to make meaningful predictions and practical use of such data. An important practical application of this convergence is providing a scientific framework for regulatory guidance involving characterization of highly heterogeneous molecules that make up active pharmaceutical ingredients of complex drugs. This chapter provides a perspective on developing such a framework for characterization of highly heterogeneous drugs from the standpoint of demonstrating sameness or equivalence with a reference product. Using examples of recently approved generic versions of complex drugs, this chapter presents the concept of "test-points" (terminology used in design of integrated circuit boards) that capture critical relationships between structural attributes of the heterogeneous molecules, the process steps involved in generating the drug product, and the starting reagents or materials.

Keywords Test-point · Big-data · Orthogonal · Heterogeneity · Regulatory · Drug

18.1 Introduction

Recently, the life sciences field has been undergoing a major transformation, both from the standpoint of broadening the context (one molecule, one cell to multicellular systems) and the development of novel, high-resolution tools (analytical, biochemical, biological assays, etc.). Because of this transformation, there has been rapid

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R. Sasisekharan et al. (eds.), *The Science and Regulations of Naturally Derived Complex Drugs*, AAPS Advances in the Pharmaceutical Sciences Series 32,
https://doi.org/10.1007/978-3-030-11751-1_18

expansion of datasets through advances in -omics technologies (high-throughput analysis at a systems level), including the availability of so-called big data (next-generation sequencing, etc.). This unprecedented volume of data and the underlying complexities for data analytics warrants the need to develop concepts or frameworks to enable practical interpretation and utilization of this information. Paralleling the transformation in the life sciences, there has been rapid advances in the development of analytical technologies for characterization of highly heterogeneous molecules both in the context of their physiological milieu and in defining the active ingredients of complex drug products. For example, there have been powerful advances in chromatography (both in column and flow design), mass spectrometry (toward achieving higher resolution and sensitivity), and nuclear magnetic resonance technologies (improved sensitivity and accuracy), resulting in the production of increasing data on the various structural attributes of complex products at the molecular level. Such highly heterogeneous molecules provide meaningful examples to illustrate the need for developing a scientific strategy or approach for characterizing and comparing these molecules using the large amount of resulting datasets for various applications.

The need for a strategy for comparative characterization is in (1) enabling the discovery and development of new products from natural sources or from complex manufacturing processes/starting materials; and (2) guiding scientific and regulatory evaluation of generic versions of complex drugs or biosimilars. For instance, generic drug products or biosimilars containing a mixture of heterogeneous molecules are compared to innovator drugs to aid product development and eliminate or reduce the need for clinical studies. While classes of heterogeneous drugs may require clinical testing and comparison to a reference product to ensure comparability, rapid developments in analytical technologies, including both product characterization [1, 2] and measurement of biological responses [3], enables a scientifically driven development of structure–activity relationships and comparison of materials to one another, especially when one can be used as a reference.

This chapter provides a perspective on strategies for characterization of heterogeneous drugs from the standpoint of demonstrating sameness or equivalence with a reference product. Specifically, this chapter presents the concept of integration and cross-correlation of datasets from multiple orthogonal techniques to define critical relationships between structural attributes of the heterogeneous molecules, the process steps involved in generating the drug product and the starting reagents or materials. These critical relationships are often nonlinear and can be considered analogous to “test-points” in an input/output circuit board. The concepts are illustrated with the help of specific examples including polysaccharide (enoxaparin) and polypeptide mixtures (glatiramer acetate) given that the US FDA has been closely involved in framing the scientific basis for approving generic versions of these drugs [1].

18.2 Heterogeneity in Complex Drugs

While the term complex mixture is often used generically to describe active pharmaceutical ingredients (APIs) that do not conform to a simple chemical formula, there are likely multiple categories within this catch-all collection. First, a distinction can be made between complex formulations and complex APIs. The former category includes lipid or nanoparticle-based products, such as amphotericin B lipid complex and nanoparticle albumin-bound paclitaxel. These products will not be discussed further; however, concepts expressed here are undoubtedly useful to characterize and define these APIs. Additionally, within the classification of complex APIs, a distinction can be drawn whether the active material is primarily protein or polypeptide/polysaccharide.

Polypeptide, polynucleotide, and/or polysaccharide mixtures are known to possess biological, and in some cases, therapeutic function. Often these mixtures are biologically derived, such as through recombinant DNA technology, as is the case for recombinant proteins, enzymes, and antibodies (e.g., epoetin alpha, laronidase, and trastuzumab). In other instances, these products are semi- or fully synthetic, for example, in the case of pentosan polysulfate or hydroxyethyl starch. This chapter, and its examples, will largely focus on the second class of complex APIs, viz., polypeptide/polysaccharide-based APIs.

Heterogeneity in the API of complex drugs is introduced at two levels. The first level is the extraction of the starting material from a plant or animal biological source. In the case of polysaccharide-based APIs such as pentosan polysulfate or enoxaparin, the starting material heterogeneity arises from a combination of a non-template-driven biosynthetic machinery and the processes used for their extraction from their representative biological sources. On the other hand, in the case of synthetic polypeptide mixtures such as glatiramer acetate, there is minimal heterogeneity in the starting materials given that they are derivatives of defined amino acids. Given biosynthetic machinery governs heterogeneity of biological source-derived starting materials, there are inherent patterns that provide important test-points to connect starting material to final (polysaccharide) API. Initially, the process used to extract starting material from its biological source introduces structural heterogeneity, such as affecting chain length, overall chain properties, and introducing alterations in molecular structure. The second level of heterogeneity is introduced during the upstream and/or downstream processes used in the generation or manufacture of API from the appropriate starting material. In the case of enoxaparin, the critical process step of depolymerization (which reduces the molecular weight/average chain length) introduces additional heterogeneity in terms of modification to the non-reducing and reducing ends. In the case of glatiramer acetate, the heterogeneity in the polypetide mixture is completely determined by the synthetic steps starting from initial molar ratios of the starting reagents and the nature of the initiating reagent.

18.3 Integrated Approach for Comparative Characterization

From the above examples, it is evident that the heterogeneity of polysaccharide or polypeptide-based APIs is such that there is no single chain with a "sequence," in the traditional sense present at substantial levels, certainly not representing even one percent of the entire mixture. As such, unlike with traditional APIs, and even glycoproteins, there arises a definitional question of *what constitutes the active ingredient and how does one compare a potential generic product to a reference*?

To answer this, overarching question requires (1) developing an analytical framework to be able to measure quantitatively multiple attributes at different subpopulation distributions (i.e., "layers") based on the understanding of the sources of heterogeneity and the need for orthogonal measurements of these attributes on each layer of a complex API; (2) measure the variance in these attributes across multiple lots of material and cross-validating these attributes derived from orthogonal measurements to ensure sufficient characterization (e.g., no missed attributes owing to method sensitivity); and (3) developing a predictive framework that captures key correlations between API, staring material, and process steps to provide critical test-points from standpoint of comparison between generic and reference product (Fig. 18.1). In terms of developing an analytical framework, many high-resolution techniques have been developed to examine complex mixture products, including glycoproteins [2] and polysaccharides [3]. Of special note are advances in mass spectrometry and nuclear magnetic resonance. Essential in this entire characterization program is the demonstration that the methods employed are:

- *accurate*, able to correctly identify and quantify particular molecular attributes;
- *sensitive,* able to detect entities at low levels;
- *specific*, able to distinguish materials that are not the same as the reference product.

Importantly, this framework, especially the last bullet encompasses aspects of quality by design [4], namely, the ability for the analytical framework to detect differences within molecular attributes upon perturbation of the process used to create the material.

In addition to specific analytical techniques, and their outputs, mathematical integration of analytical data provides a powerful method to demonstrate aspects of the above three bullets. Notably, as the features of a complex API are quantified using analytical methods, the number of possible molecular species grows rapidly [5]. Thus, while a single analytical measurement, no matter how data rich, might not greatly restrict the solution space sufficiently, a combination of well-selected measurements can be surprisingly restrictive when formalized [6]. Use of a mathematical framework to catalogue and integrate results from analytical datasets has most often been used in the context of compilation of -omics data and pathway analysis or discerning patterns/attributes. Several successful approaches have been developed to integrate results from analytical measurements of biological mixtures, including iterative removal of potential solutions that do not solve the analytical measurements

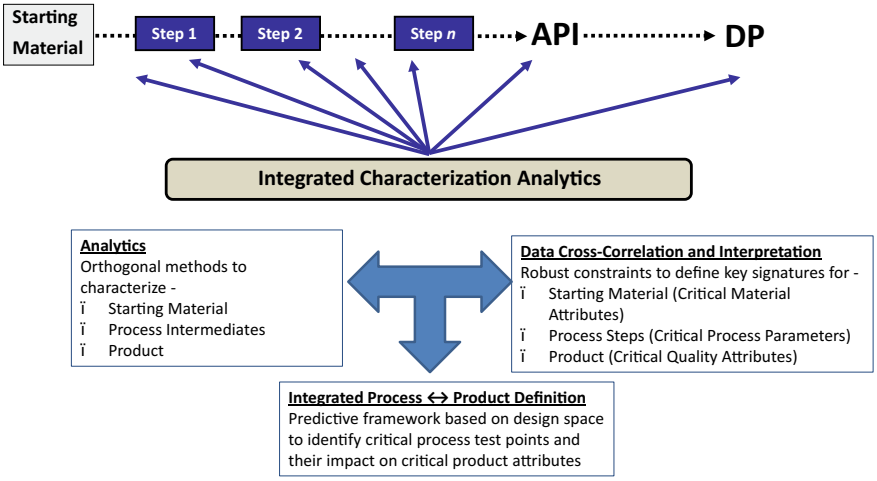


Fig. 18.1 *Integrated approach for characterization of highly heterogenous molecules.* The source of heterogeneity of the molecule is impacted by the inherent heterogeneity of the staring material and the key steps in the process chemistry on the starting material to give the final drug substance (or active pharmaceutical ingredient). The analytical framework comprises of a layered approach to heterogeneity that involves using orthogonal measurements on the starting material, process intermediates and final API and correlations of the attributes from these measurements across the layers to provide a robust and deep characterization of the heterogeneous mixture. The final aspect of characterization involves defining key relationships between process and product attributes which permit establishing critical test-points for defining the mixture and to demonstrate sameness between generic and reference product

[7] as well as use of advanced metrics, such as constrained optimization and linear programming, to yield bounds on the molecular abundances of molecular species [6].

Therefore, taken together—analytics, data cross-correlation, predictive test-points for process to product relationships, and biological testing—provide the basis of addressing the definitional question of what constitutes a *complex mixture* and therefore allows one to compare materials, either lot-to-lot and/or to a reference.

18.4 Examples to Illustrate Integrated Approach

18.4.1 Enoxaparin Sodium

The starting material for generating enoxaparin is USP grade unfractionated heparin. Heparin is a well-known anticoagulant drug and is extensively used in medical practice [8]. Heparin is isolated from animal organs, predominantly porcine intestinal mucosa, and goes through an extensive process of purification before it can be

used for pharmaceutical purposes [9, 10]. The molecular basis for the anticoagulant function of heparin was elucidated in the early 1980s when a distinct pentasaccharide sequence within heparin chains was identified as being crucial for binding and activating antithrombin, leading to accelerated inhibition of the coagulation cascade [11, 12].

In taking action on requests to market generic versions of enoxaparin sodium within the United States market, the FDA published a comprehensive document describing the technical and regulatory basis of approval (Docket No. FDA-2003-P-0273). In this response, the FDA constructed a five-point set of criteria for establishing sameness or "standards for identity" (i.e., comparison of the generic drug with the RLD) [13]. This document answered, from a scientific and regulatory prospective, important definitional questions including how does one robustly compare a product to a reference and establish "active pharmaceutical ingredient sameness." The following recapitulates these criteria based on the integrated approach described in this chapter.

18.4.1.1 Sources of Heterogeneity in Enoxaparin

As stated earlier, the heterogeneity in enoxaparin related to the unfractionated heparin starting material is governed by the biosynthesis of heparin in porcine mucosa and the manufacturing processes to isolate and purify heparin. Examples of how extraction process of heparin impacts its structure include the formation of 2,3-epoxides, galacturonic acid, and cleavage of the linkage region (Fig. 18.2). For example, oxidation by potassium permanganate is used by some manufacturers to remove the core protein from the heparin product. This reaction alters the reducing end of heparin through several chemical side-reactions, leading to unnatural saccharide residues [14]. Other structural changes in heparin (i.e., *O*-acetylation and desulfation) resulting from manufacturing processes have also been reported [15, 16]. Additionally, several groups have identified the fact that addition of oxidizing agents, such as peracetic acid, can result in transient *O*-acetylation.

Furthermore, during the depolymerization process, the introduction of a benzyl ester onto the carboxylate of uronic acids of heparin forms sites of potential cleavage within the heparin chain (Fig. 18.3). Addition of base results in two competing reactions: proton abstraction with cleavage of the glycosidic linkage as well as hydrolysis of the ester. In addition, several minor reactions occur, including formation of 1,6 anhydro rings at the reducing ends of some chains, peeling reactions, and the formation of additional epoxide moieties. Importantly, these modifications provide important imprints of the process. Following the same inherent chemistry, i.e., aqueous cleavage of a heparin benzyl ester with a defined range of esterification, and ensuring these process signatures are present at the same levels within the comparator product as in the reference product ensures that the range of potential variation in the API is restricted and controlled.

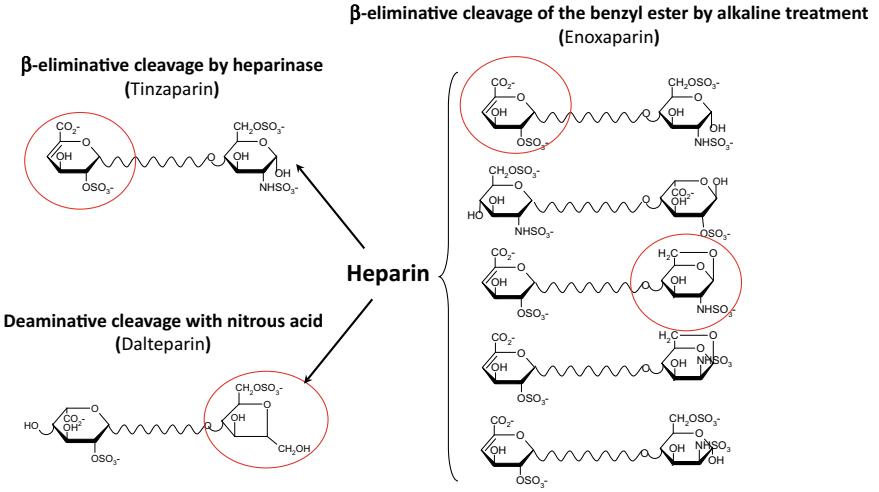


Fig. 18.2 *Low-molecular-weight heparins are derived from heparin through controlled depolymerization.* Three LMWH products have been on the US market—tinzaparin, dalteparin, and enoxaparin. Tinzaparin is generated through enzymatic cleavage of heparin through the use of heparinase I. Conversely, dalteparin and enoxaparin result from chemical cleavage of heparin. Dalteparin is generated through the use of nitrous acid whereas enoxaparin is generated through base cleavage of a heparin benzyl Easter. Each process results in modifications to heparin, mostly at the ends (reducing and non-reducing) of the newly formed chains, which is unique to the depolymerization process. These modifications are shown with a red circle

18.4.1.2 Analytical Framework for Enoxaparin

The different layers of the heterogeneous polysaccharide mixture in enoxaparin can be described using orthogonal approaches:

(i) *High-level physiochemical properties*: Such properties include, but are not limited to, molecular weight and molecular-weight distribution, sulfate-to-carboxylate ratio, and features in the 1H and 13C one-dimensional nuclear magnetic resonance (NMR) spectrum. Importantly, these attributes have been shown to capture key signatures associated with heterogeneity in the source heparin starting material and mode of depolymerization.

(ii) *Building blocks*: Equivalence in compositional analysis between a reference and a comparator can be established by demonstrating equivalence in the identity and quantitative levels of disaccharide and other small oligosaccharide units which constitute the oligosaccharide chains of enoxaparin sodium. As noted in Table 18.1, multiple methods, including enzymatic treatment followed by HPLC or CE and two-dimensional NMR can be used to establish the identity and quantity of building blocks within enoxaparin sodium and to compare materials to one another.

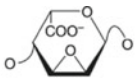
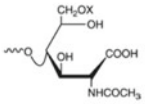
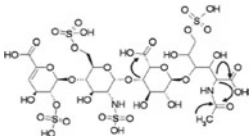
Modification	Structure	Formation
2,3-epoxide		• Based-catalyzed attack after removal of a 2-*O* sulfate • Can ring open to galacturonic acid
N-acetyl glucoamine oxidatation		• Oxidation of reducing end occurs with addition of potassium permanganate
Tetrasaccharide		• Potassium permanganate-derived (not present when hydrogen peroxide is used)
O-acetylation	None disclosed	• Transient O-acetylation from addition of peracetic acid

Fig. 18.3 *Monosaccharide modifications that can occur in heparin as part of its manufacturing process.* In the isolation and purification of heparin from porcine intestinal mucosa, several heat, acid, and/or base steps are used to degrade other biological material, such as proteins and nucleic acids, followed by salt/organic-based precipitation steps, taking advantage of heparin's pronounce polyelectrolytic nature. These steps can introduce modifications into the heparin backbone

Table 18.1 Selected analytical methods used to characterize heparin/LMWH

Method	Use	References
One-dimensional NMR	Structural profiling	[29, 30]
Two-dimensional NMR	Monosaccharide identification and quantification	[31]
Enzymatic digestion and liquid chromatography/capillary electrophoresis	Quantification of constituent parts	[28, 32]
LC-MS	Structural characterization of components	[33–35]
Partial digestion	Mapping/quantification of fragments	[36]
Gel permeation chromatography-MS	Analysis of size-separated fractions	[37]
Sequencing chains	Assignment of structure to chains within mixture	[7, 38]

(iii) *Higher-order oligomeric species*: In addition to compositional analysis, it is important to conduct additional analyses to confirm the distribution of building-block units in the oligosaccharide chains. This information can be obtained, for example, by fragment mapping which is analogous to a tryptic mapping for proteins. Partial digestion of enoxaparin sodium and analysis of the resulting products provide a signature of recurring oligosaccharide sequences unique to enoxaparin sodium, and thus gives evidence that comparator possesses the same recurring oligosaccharide sequence segments as those in the reference product.

(iv) *Measurement of structural attributes in sub-populations based on size fractions*: Dividing the mixture based on size fractions permits the measurement of attributes such as building blocks and oligomer fragments within each fraction. These measurements enrich the analytical datasets and provide additional constraints for cross-correlation of these measurements to those obtained for the entire mixture. Fractionation also facilitates direct sequencing of a subset of oligosaccharide chains that provides more granularity from the standpoint of defining the mixture and for comparison with reference product. As shown in Table 18.1, several sequencing technologies have been developed to sequence oligosaccharide chains.

18.4.1.3 Test-Points for Definition and Comparative Characterization of Enoxaparin

The aforementioned analytical framework would provide several critical attributes of the reference product at different layers of heterogeneity and their cross-correlations across the layers. For the specific attributes in the API (and process intermediates during development of manufacturing process) that correlate with predicted heterogeneity as it is impacted by biosynthesis, heparin extraction, and depolymerizaton processes would provide the key test-points for linking product with starting material and process steps. From a biosynthesis standpoint, it is known that specific patterns of repeat units of 2-*O* sulfated iduronic acid linked α1→4 to a *N*- and 6-*O* sulfated glucosamine are observed in the starting material and that its levels read through to the API. Additionally, a specific pattern of glucuronic acid β1-4-linked to *N*-, 3-*O* and 6-*O* triple sulfated glucosamine which is a part of the active pentasaccharide sequence, responsible for the API's anticoagulant activity, can be correlated in terms of relative abundance between staring material and API. The signatures associated with depolymerization including molecular-weight distribution differences and specific modifications in the non-reducing and reducing end (such as 1,6-andhyro structure) can be used to connect starting material and process step with the API.

The aforementioned approach provides a scientific basis not only for comparative characterization of generic and reference complex drug but it also provides a rational way for developing chemistry and manufacturing process for generic enoxaparin. Another important aspect of characterization is to assess the activity of API in biochemical, biological and in vivo assays, a topic which is covered elsewhere.

One post-script to the establishment of this framework is the evolving specification for heparin in the wake of the heparin contamination crisis. In early 2008, a research team, including academia, the FDA, and industry identified a contaminant, oversulfated chondroitin sulfate (OSCS) [19], in lots of heparin. OSCS was found to be a powerful activator of the contact system [20]. Consequently, the USP in collaboration with FDA initiated a rapid revision of the existing control measures for heparin. Initial changes, implemented in the United States after only three months, included the addition of critical identity tests, employing capillary electrophoresis, and proton nuclear magnetic resonance (1H-NMR). Since this time, additional changes have been incorporated, including addition of a test for molecular-weight distribution, assays for anti-Xa and anti-IIa activity, and several tests to measure impurities, including proteins and nucleic acids, if present. This comprehensive overhaul of quality measures for heparin enhanced the quality of this drug by improved detection of contamination, reducing impurity levels, and establishing identity tests that are sensitive to the structure and composition of the drug [21]. Notable reduction in the impurity level of dermatan sulfate in heparin products was reported in those samples collected in 2009, after the revised heparin monograph was introduced [21]. One consequence of the routine introduction of enhanced analytics, especially NMR (see above) and of likely improvements to manufacturing steps in the creation of heparin has been the alteration in the fine structure of heparin. This in turn is anticipated to affect the fine structure of enoxaparin sodium, resulting in some level of manufacturing drift in lots over time. Because of this, high-resolution analytics, such as NMR and MS, which have become widespread in laboratories, likely should be incorporated into periodic lot analysis and potentially lot release.

18.4.2 Glatiramer Acetate

Glatiramer acetate, in contrast to enoxaparin sodium, is not biologically derived, but rather is created from a process of controlled polymerization of four activated NCAs—those of alanine, lysine, glutamic acid, and tyrosine [22]. The formation of polypeptides from NCAs involves three processes: *initiation*, *propagation,* and *termination*. Initiation and propagation involve the addition of a nucleophile to the C-5 of the NCA ring (Fig. 18.4). Since this reaction results in opening of the NCA ring, NCA polymerization falls into a general category of polymerization called "Ring-Opening Polymerization" (ROP) [23]. For amines that are nucleophilic, e.g., primary and secondary amines such as diethylamine, the reaction proceeds by nucleophilic attack of an amine on an NCA for each initiation and propagation step. The initiation reaction of DEA with an NCA and subsequent propagation reactions to elongate polypeptide chains are shown in Fig. 18.4.

In addition to chain formation through reaction of activated NCAs, additional steps are used to produce glatiramer acetate (Fig. 18.5). Polymerized chains of alanine, lysine, glutamic acid, and tyrosine are then subjected to acid-mediated depolymerization and deprotection of the benzyl ester protecting group on glutamic acid

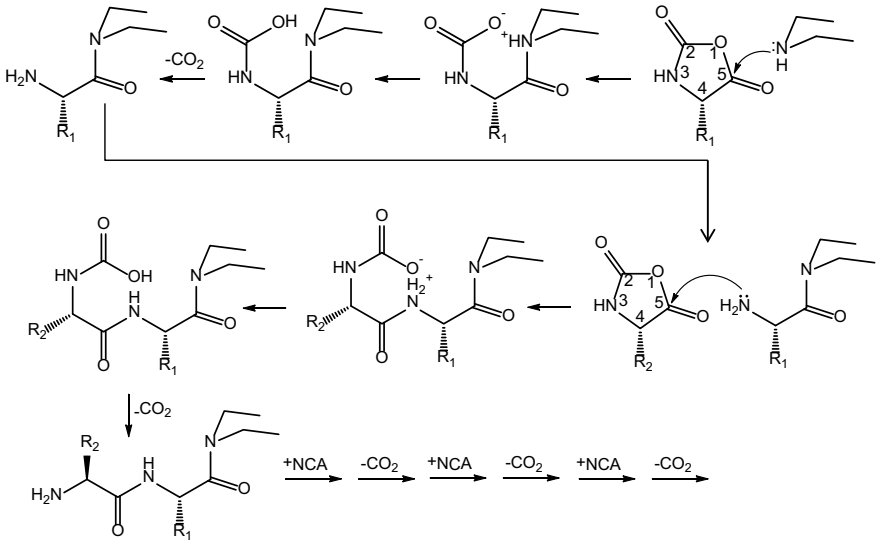


Fig. 18.4 *Chain initiation and initial propagation in the formation of glatiramer actetate.* NCAs in dioxane or other polar, aprotic solvent are activated through nucleophilic attack by diethylamine to form an amide, which marks the C-terminal end of a chain. This ring-opened NCA exposes an amine that can attack additional NCAs, resulting in chain polymerization

residues. Because of chain scission, the average molar mass of chains is reduced substantially; accordingly, the number of chains increases. Then, treatment with a base, such as piperidine, removes the trifluoracetamide (TFA) protection group from lysine residues. Finally, filtration and isolation of peptide material is used to remove process reagents, impurities and to convert the lysine amine residues to the acetate salt.

18.4.2.1 Sources of Heterogeneity in Glatiramer Acetate

The starting reagents including four NCAs, initiator, and solvent are chemicals that are typically very pure and well defined. The heterogeneity of polypeptide mixture in API is primarily governed by the polymerization, depolymerization, and purification process steps. Because chain initiation, chain propagation, and chain scission are governed by the relative kinetics of the reactions, glatiramer acetate is a mixture of high complexity, consisting of a collection of diverse peptide sequences each of which are present at very low abundance. As a result of this complexity, representing glatiramer acetate by intact chain sequences and their relative abundances is not meaningful, both because no single chain will represent a meaningful portion of the mixture and because no two lots of glatiramer acetate will contain the same mixture of chains.

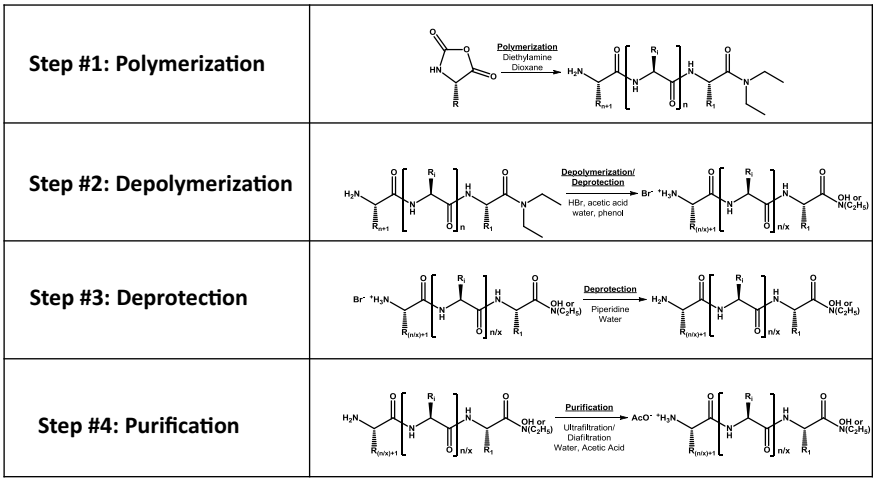


Fig. 18.5 *Steps in the production of glatiramer acetate*. The overall process to produce copolymers, such as glatiramer acetate, can be divided into several distinct steps. In Step #1, the chain composition is locked in through the inherent polymerization kinetics of the NCAs. Step #2, depolymerization, lowers the molecular weight and creates the bulk of the chains present in the final material. Amino acids with functional groups on their side chains, such as glutamate and lysine, must be protected during step #1. Deprotection, during step #3, removes the protecting groups. API is isolated in the final step

Therefore, heterogeneity in glatiramer acetate is best captured by understanding shifts in relative abundances of NCA during chain initiation and propagation (referred to as propagational shift) and structural attributes of the polypeptide that correlate with these events [24]. Even though characterization of intact polypeptide chains would not provide a meaningful description, the imprint of propagational shift as local sequence patterns within the polypetide chains (analogous to biosynthesis imprint of specific repeat units on polysaccharide mixture of enoxaparin) can be captured.

The heterogeneity associated with a depolymerization event would again be governed by the relative kinetic rates of acid hydrolysis cleavage of the different peptide bonds and the effects of these kinetic events manifest as structural attributes of the N- and C-termini of the polypeptides after depolymerization.

18.4.2.2 Analytical Framework for Glatiramer Acetate

(i) *High level physiochemical properties of mixture*: These include, but are not limited to, molecular-weight distribution, spectroscopic fingerprints

(ii) *Building blocks of the peptide*: Building blocks of polypeptide mixture, include, but are not limited to, relative abundance of amino acids (which would be very similar to initial ratios of NCAs) and specific signatures associated with chain initiation such as ratio of DEA-capped C-terminal amino acids) and

depolymerization such as ratio of free (or uncapped) C-termini to DEA-capped C-termini

(iii) *Higher-order attributes*: Higher-order attributes of the mixture capture key signatures of chain propagation and depolymerization events. Characterization of N-terminal amino acid distribution through repeated cycles of N-terminal sequencing provides key information on depolymerization. Distinct mass ions (and possible annotation of sequence motifs) from LC-MS analyses that can discriminate API of reference product with other generics and negative controls [25, 26]. The ability of these local sequence motifs to discriminate RLD is an indirect imprint on the effect of relative rates of NCA incorporation (or propagational shift) into the chains

(iv) *Analysis of sub-populations of mixture*: These include fractionation of chains based on size or charge and analysis of the attributes for each fraction would provide key signatures of sub-populations that capture chain initiation, propagation, and depolymerization events that can be correlated with the properties of the entire mixture.

18.4.2.3 Test-Points for Definition of Glatiramer Acetate

As stated earlier, heterogeneity in glatiramer acetate primarily arises from the synthetic reaction scheme involving chain initiation, polymerization or propagation, depolymerization, and purification (Fig. 18.5). Importantly, in the polymerization step, due to the differential reaction kinetics, the different NCAs are consumed at different rates, and consequently, the proportion of each NCA changes as a function of reaction time. For example, the proportion of alanine incorporated into the polymer decreases as the reaction proceeds, since the alanine-NCA has the fastest propagation rate constant. Conversely, the proportion of tyrosine-NCA increases as the reaction proceeds, since it has the slowest propagation rate constants. Each of these steps can be modeled mathematically as kinetic events by independently determining rate constants for initiation, polymerization (which in turn can be simply defined as homo or hetero polymerization), and depolymerization events. The model can also be set up to simulate the chains to provide a means for correlating with key signatures (measured using the analytical framework) associated with chain initiation, propagational shift, and depolymerization events. In this fashion, the test-points here would provide key link between chemistry (e.g., kinetically controlled NCA polymerization initiated by diethylamine), analytics, chemical knowledge with the help of a kinetics-based mathematical model of the process steps to provide a robust framework for defining glatiramer acetate, and assessment of sameness.

18.5 Mathematical Models for Characterization of Heterogeneous Molecules

Typically, mathematical resolution of mixtures is achieved using chemometric type analyses. Such chemometric approaches have been used to discriminate LMWH generic and innovator lots [6, 27]. One such approach analyzed the relative abundance of disaccharide building blocks in a fully depolymerized LMWH mixture using HPLC [27]. Using these measurements, partial least squares discriminant analysis (PLS-DA), for example, can be used to classify the samples based on manufacturer and to examine the differences in composition that make such a classification possible. PLS-DA is a highly used statistical method in chemometrics when trying to discriminate a categorical variable (e.g., LMWH manufacturer) using correlated and non-normally distributed independent variables [28]. It was shown that specific disaccharide building blocks are associated with the different manufacturers, and produce a model that could theoretically classify future examples by manufacturer. PLS-DA is an example of a method offers many advantages, including variable importance projections for feature selection, score-score plots for low-dimensional visualization, and loading plots to visualize the features most closely associated with the categorical variables. However, this type of model is useful for characterizing the *differences* between two or more mixtures, it does not provide a robust definition of the complex active ingredient.

Alternatively, a mathematical approach that formalizes the three components of integrated comparative characterization highlighted above offers several advantages. First, a refined model can provide maximal information by combining a priori knowledge of the system with the analytical measurements, as is done, for example, in [6]. In addition, such a model may be constructed to estimate confidence intervals (e.g., measure the variance) not only for individual measurements, but also the characteristics of interest by aggregating data across measurements. Suppose a quantity can be separately measured by both LC-MS and NMR, then the model can combine these measurements to assess the mean value and variance of the underlying quantity while minimizing bias that is inherent in the use of a single measurement. Finally, the process of formalizing the model requires a more explicit examination of assumptions and measurement limitations. This modeling framework is highly flexible and its precise application is problem specific. However, there is a general framework which typically serves as a base for formulating the model.

Constrained optimization provides a natural starting point for a mathematical model for integrated comparative characterization. Broadly, the system consists of the following: (1) Observations obtained by the measurements O. (2) A model M relating a set of parameters θ to the observations. (3) A "Loss Function," L which relates the model to the observations using some criterion of optimality. (4) A set of constraints g and h below:

$$\min_{\theta} L(M(\theta), O) \text{ s.t.} \tag{18.1}$$

$$g(\theta) = \mu;\ h(\theta) \leq \vartheta \tag{18.2}$$

This is the most general formulation of a typical constrained optimization, but the exact setup may look different depending on the problem. The utility of the framework is that it allows a flexible description of the system under consideration while also formalizing any assumptions and a priori information.

To make the equations more concrete, consider a polymerization reaction. The parameters here could include kinetic rate constants, reaction time, concentration of starting materials, etc. Some of these rate constants may be known, in which case terms in $g(\theta) = \mu$ would reflect the knowledge of these values. Others may be bounded, in which case they would be incorporated in $h(\theta) \leq \vartheta$. Still, other parameters may only appear in (18.1), because there is no a priori knowledge of their values. As part of the model we may want to predict the moments of the molecular-weight distribution using the input parameters. M would map the parameters θ to the moments of the MWD (O), and the loss function would define a metric for how close the model is to the observed values when using any specific set of parameters. Various algorithms exist to optimize the values of the unknown parameters, and others can provide bounds and potentially confidence intervals.

A recent study [6] employed this constrained optimization framework to examine the polysaccharide mixture present in bovine kidney heparan sulfate (BKHS). To accomplish this goal, the number of chemical modifications to the saccharide backbone was reduced from 13 to 2 to make the mathematical formulation a more tractable problem. Using a combination of orthogonal measurements from NMR and liquid chromatography-mass spectrometry (LC-MS) of BKHS digested with heparinases as constraints, a non-homogeneous Markov chain model is used to explain the length distributions obtained when cleaving BKHS with heparinases I and III based on the sequence of modifications in the original BKHS chain. This model requires a parameter calculated from the different cleavage yields of heparinases I and III on the sulfated and unsulfated building blocks, which is calculated through the use of both 1H NMR and LC-MS. In this case, using multiple technologies provides a higher level of certainty in parameter estimation. Their generative model then predicts the length distributions observed in BKHS digested by heparinase I or III using parameters describing the probability of sulfation along BKHS chains. This system highlights the need to integrate orthogonal measurements as constraints to robustly define a mixture using mathematical framework.

18.6 Conclusions

The resolving power and amount of information available from analytical approaches have and continue to increase dramatically, due to both improvements in instrumentation and computation. These advances lead to large volumes of datasets on complex heterogeneous molecules in drug products and thereby require development or adaptation of concepts and strategies in big data analysis space to enable practical use of

these data such as in making scientifically guided regulatory decisions for approval of such complex drugs. By employing a mathematical framework to integrate attributes measured from orthogonal analytical data, it is possible to achieve a robust product definition, at a molecular level, and completing robust comparisons of any manufacturer's materials to a reference. Such approaches should only become more powerful as our experience, and technology, improves.

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